# The PairPlot addon

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This package produces grids of subplots depicting pairwise relationships in a dataset. This package maps each variable in a list onto a column and row in a grid of multiple axes.

It can also represent an additional level of conditionalization with the 'factor' argument (a series) which plots different subsets of data in different colors and/or point/circle types. This is used to resolve elements on a third dimension.

Please report bugs or comments on the gretl mailing list, report an issue on github ( https://github.com/atecon/PairPlot/issues ) or write to atecon@posteo.de.

## 1 Usage via the GUI

The function `PairPlot()` can be accessed through the GUI. The menu entry is "View->Graph specified vars->Pair Plot". However, via scripting the user can control many more details.

## 2 Public function

```
PairPlot(const list L, const series factor[null], bundle opts[null])
```

Main function for creating pair-plots.

**Parameters**

- `L`: list, Series to plot against each other
- `factor`: series, Categorical (numeric or string-valued) variable for mapping plot aspects to different colors and/or points/ circles (optional, default: null). The factor series should have at most 16 distinct values. This series must be of type 'discrete'. To set a series to discrete, use the command `setinfo <series name> --discrete`.
- `opts`: bundle, Optional for passing additional parameters. See below for explanation (default: null)

**Returns**

An integer of value '0' (FALSE) if no error occurs, otherwise 1 (TRUE).

**Options passed to the PairPlot() function by means of the 'opts' bundle**

The user can control the following aspects by adding the respective parameter to the 'opts' bundle.

- `centroid`: string, Draw a symbol highlighting the centroid of the 2-dimensional data. Possible options are `mean`, `median` and `none` (default: "none"). If a `factor` series is passed, the centroid for each distinct factor is shown.
- `centroid_label`: bool, Show the label of the centroid statistics if TRUE, otherwise do not show if FALSE (default: FALSE).

- `centroid_linewidth`: scalar, Width of the line (default: 2).
- `centroid_pointsize`: scalar, Size of the point (default: 2).
- `filename`: string, Path plus filename plus file type. (optional, default: `display` which draws the resulting plot directly at the screen). Currently figures can only be stored in **png** format.
- `fontsize`: int, Control the font size of the labels (default: 16) .
- `grid`: bool, Draw a grid in the background if TRUE (=1) (default: FALSE)
- `height`: scalar, Height of the canvas plot (default: 600, min. size: 100).
- `title`: string: set an overall title for the plot(s).
- `key`: bool, If the `factor` series is provided, a legend shows the color and point pattern for each distinct value of the `factor` variable. Default: 1 (TRUE).
- `key_fontsize` int, Control the font size for the key. Default: 14
- `key_position`: string, Controls the position of the legend in each subplot (use standard gnuplot options). default: "top left".
- `offset_level`: scalar, Provides a mechanism to put a boundary around the data inside of an auto-scaled graph. For details see: https://gnuplot.sourceforge.net/docs_6.0/loc14594.html , Default: 0.02.
- `tmargin`: scalar, Top margin of the plot in lines. Positive values move the
- `bmargin`: scalar, Bottom margin of the plot in lines.
- `lmargin`: scalar, Left margin of the plot in lines.
- `rmargin`: scalar, Right margin of the plot in lines.
- `pointsize`: int, Control the size of points (default: 2). No support for circles.
- `pointtype`: matrix, Control the type of points. Can be either an integer entry such that points associated to different levels of `factor` are of the same type. One may also provide a vector of integers for each distinct value of the `factor` series. Default: 7. See also: https://i.imgur.com/xNZa7Rz.png
- `tics`: bool, Print x-tics and y-tics if 1 (TRUE, default), or not if 0 (FALSE).
- `tics_fontsize` int, Control the font size of the x-y tics. Default: 12
- `transparency_level`: int, Controls the transparency level of points and circles (between 0 (no transparency) and 255 (full transparency)). Default: 90.
- `type`: string, Select the type of plot (default: "triangle"). The following types are supported:
- **(i)** "matrix": Scatterplot matrix which draws a scatter plot between all combinations of variables included in list `L`.
- **(ii)** "triangle": Lower triangular version of the scatterplot "matrix" type.
- **(iii)** "row": Plot the first series element in list `L` in a single row against all remaining list members.
- **(iv)** "column": Works like type "row" but subplots are arranged in a single column instead.
- `use_circles`: bool, Draw circles instead of points if set to 1 (TRUE), default: 0 (FALSE).
- `width`: scalar, Width of the canvas plot (default: 900, min. size: 100).

- `xlabel_offset_horizontal`: scalar, Horizontal offset for the X-axis label (default: 0). Negative value move the label to the left, and vice versa.
- `xlabel_offset_vertical`: scalar, Vertical offset for the X-axis label (default: 0). Negative values move the label downwards, and vice versa.
- `ylabel_offset_horizontal`: scalar, Horizontal offset for the Y-axis label (default: 0). Negative value move the label to the left, and vice versa.
- `ylabel_offset_vertical`: scalar, Vertical offset for the Y-axis label (default: 0). Negative values move the label downwards, and vice versa.

## 3 Changelog

- **October 2026**

  Switch help to LaTeX (pdf) format.

- **v1.1 (July 2025)**
  - For triangle-type of plot: improved x-/y-label positioning
  - Make margins of the plot configurable via `tmargin`, `bmargin`, `lmargin` and `rmargin` parameters.
  - Added `xlabel_offset_horizontal`, `xlabel_offset_vertical`, `ylabel_offset_horizontal` and `ylabel_offset_vertical` parameter to control X-axis and Y-axis label offset.
- **v1.0 (February 2025)**
  - Bugfix: Catch error in case size of width or height is less than 100.
  - Bugfix: Sample 2 not working before as reference to dataset `iris.gdt` was missing.
  - Ensure that the `factor` series is of type 'discrete'. If not, an error message is thrown.
  - Throw an error if the `factor` series has more than 16 distinct values.
- **v0.99 (July 2024)**
  - Add new parameter `title` for setting an overall title
  - Internal refactoring: Switch to gretl's built-in gridplot apparatus which means that all the graphics file formats supported by gretl can be used.
  - It also means that usage via gretlcli in "display" mode works.
  - In case the user's specification calls for just a single plot, no "multiplot" is created.
  - The graphical interface is enhanced, with more options.
- **v0.98 (April 2024)**
  - Make width, height, font size and point size a function of the number of variables for better readability in case of many variables.
  - Set parameter `transparency` per default to 90.
  - Minor update of the sample script
- **v0.97 (February 2024)**
  - Error in case of CLI-mode if a plot is expected to appear on the screen (filename="display")
- **v0.96 (February 2024)**
  - Set all terminals to 'noenhanced' mode to respect eventual underscores in variable names

  - Allow the parameters 'centroid_pointsize', 'centroid_linewidth', 'width' and 'height' to be scalar values instead of integers.

- **v0.95 (December 2023)**

  - Bugfix: Path with blanks failed under Windows sometimes.
  - Bugfix: An error occurred with negative values in some cases
  - Bugfix: An error occurred of the name of the last variable in the list starts with 'e' (or 'E')
  - Fix handling of transparency
  - Add `grid` option for (optionally) drawing a grid in the background
  - Add options `key_fontsize` and `tics_fontsize`.
  - Add support for drawing centroids of 2-dimensional space.
  - Remove the `font` option for setting the font (controlled by gretl now)
  - Increase default font size to 12
  - Switch to markdown-based help file
  - Increase minimum gretl version to 2022d

- **v0.9 (October 2022)**

  - Fix bug with `const` inheritance: its illegal for a function which receives an argument A that is declared as const to pass A on in pointer form to another function which does not promise to treat it as `const`.

- **v0.8 (January 2022)**

  - Handle eventual missing values in passed series.
  - Bugfix of gnuplot's `offset` option: More than at maximum 4 were passed in case of more than 4 features to plot.
  - Bugfix: Works now with more than 8 features.

- **v0.7 (October 2021)**

  - Enforce full sample instead of using `--replace` option
  - Changed ordering of the colors when plotting distinct factors
  - Change the color palette to "dark2"

- **v0.6 (March 2021)**

  - New API for `PairPlot()` – backward-incompatible
  - update help text
  - enforce decimal point when print numerical data to gnuplot file
  - substantial internal refactoring and clean up
  - gain of speed
  - new minimum gretl version is 2020e

- **v0.5 (Oct. 2019)**

  - Initial version